BALROG: BENCHMARKING AGENTIC LLM AND VLM REASONING ON GAMES

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ABSTRACT

Large language models (LLMs) and vision-language models (VLMs) are increasingly deployed as autonomous agents, yet existing benchmarks predominantly assess short-horizon tasks that are quickly saturating and fail to expose critical deficiencies in long-term planning, exploration, and spatial reasoning. We introduce **BALROG** (**B**enchmarking **A**gentic **L**LM and VLM **R**easoning **O**n **G**ames), a novel benchmark that aggregates six procedurally generated, long-horizon interactive game environments—BabyAI, Crafter, TextWorld, Baba Is AI, MiniHack, and NetHack—into a unified, reproducible testbed for evaluating agentic capabilities. BALROG stresses long-context decision-making in both language-only and vision-language settings, spanning tasks that range from navigation solvable in seconds to dungeon-crawling requiring years of human mastery. We evaluate leading closed- and open-source models zero-shot and find that even the best-performing language-only model achieves only 32.64% average progress, while vision-language settings often *hurt* performance relative to text-only baselines—with GPT-4o dropping from 32.34% to 22.56% when given visual input. NetHack remains largely unsolved, with the best model reaching just 1.57% progression. These results reveal fundamental gaps in long-term planning and vision-based decision-making in current LLMs and VLMs, highlighting BALROG as a rigorous benchmark to drive progress toward truly agentic AI systems. We release our benchmark, evaluation code, and a public leaderboard.

1 INTRODUCTION

The prospect of large language models (LLMs) acting as autonomous agents—perceiving environments, planning sequences of actions, and completing long-horizon tasks—has attracted significant research attention (Wang et al., 2023; Yao et al., 2024; Shinn et al., 2023; Brown, 2020). Benchmark-driven progress has been central to this effort: datasets such as MMLU (Hendrycks et al., 2020), SuperGLUE (Wang et al., 2019), SWE-Bench (Jimenez et al., 2023), and GAIA (Mialon et al., 2023) have guided improvements in language understanding, reasoning, and tool use. Yet these benchmarks share a common limitation: they assess either static knowledge or short-horizon tasks with limited interactivity, masking whether models can genuinely *act* over extended time horizons in dynamic environments.

Reinforcement learning (RL) environments have long served as challenge problems for decision-making agents (Bellemare et al., 2013; Vinyals et al., 2019; Silver et al., 2018), but adapting them to evaluate LLMs raises new questions. Can a model that *knows* how to play a game actually *do* so when actions must be chosen sequentially over thousands of steps? This knowing–doing gap (Kambhampati et al., 2024) is not revealed by standard benchmarks that prompt models for one-shot answers or short task completions.

We argue that a new class of benchmark is needed: one that tests long-horizon, interactive decision-making in procedurally generated environments where memorization is impossible, a wide range of skills is required, and difficulty is calibrated from tractable to extreme. Games are natural candidates because they are fast to simulate, richly varied, and require genuine sequential reasoning. Prior work has used individual games (NetHack (Küttler et al., 2020), Crafter (Hafner, 2021), TextWorld (Côté et al., 2019)) to evaluate specific aspects of LLM agency, but no unified benchmark spans the full spectrum of agentic skills from simple navigation to complex long-term planning.

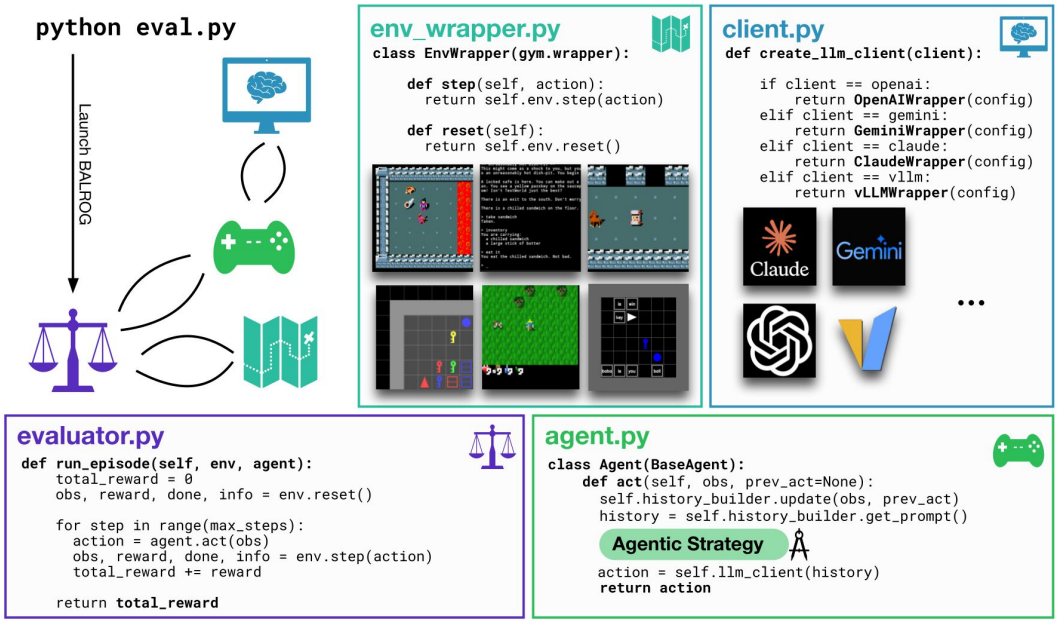

Figure 1: **An overview of the BALROG Benchmark for evaluating LLMs on long-context interactive tasks**. Models receive observations (text or image+text) and produce natural-language actions, which are executed in the environment. Results are aggregated across six game environments with a standardized 0–100 scoring scheme.

We present **BALROG** (**B**enchmarking **A**gentic **L**LM and VLM **R**easoning **O**n **G**ames), a benchmark unifying six diverse game environments with a standardized evaluation protocol. BALROG covers navigation, exploration, resource management, credit assignment, deducing environment dynamics, and long-term planning (see Table 1). Crucially, BALROG supports both language-only and vision-language evaluation modes, enabling systematic comparison of how visual context affects agentic decision-making—a question of growing importance as VLMs are increasingly deployed as agents (Rahmanzadehgervi et al., 2024).

Our main contributions are:

1. **A unified benchmark** of six long-horizon, procedurally generated game environments for evaluating LLMs and VLMs as autonomous agents, with fine-grained 0–100 scoring and a new data-informed NetHack progression metric.
2. **Comprehensive zero-shot baselines** for leading closed- and open-source models in both language-only and vision-language settings.
3. **Key empirical findings** revealing that vision input often hurts model performance, that MiniHack and NetHack are largely unsolved, and that a substantial knowing–doing gap persists even in top-tier models.
4. **An open-source toolkit** supporting rapid experimentation with diverse agentic strategies, including chain-of-thought (Wei et al., 2022), in-context learning (Agarwal et al., 2024), and reflection (Shinn et al., 2023).

2 BALROG: The Benchmark

2.1 Environment Selection and Overview

BALROG is designed around three core criteria: (1) *long-horizon* tasks requiring many sequential decisions; (2) *procedural generation* to prevent memorization and ensure robust evaluation; and (3) *breadth of skills*, spanning the range from simple navigation to complex planning requiring deep domain knowledge. Figure 1 provides an overview of the benchmark pipeline.

Table 1: **The tested skills, time horizons, and complexities of interactive decision-making tasks evaluated in BALROG**. Compared to existing benchmarks, BALROG provides infrastructure for evaluating model reasoning and decision-making on harder, longer time-horizon interactive settings. The evaluated tasks span a range of difficulties.

Skills	**BabyAI**	**TextWorld**	**Crafter**	**Baba Is AI**	**MiniHack**	**NLE**
Navigation	✔	✔	✔	✔	✔	✔
Exploration	✔	✔	✔	✔	✔	✔
Resource Management	✗	✔	✔	✗	✔	✔
Complex Credit Assignment	✗	✗	✔	✔	✔	✔
Deducing Env. Dynamics	✗	✗	✗	✔	✔	✔
Long-term Planning	✗	✗	✗	✔	✔	✔
Turns to Complete	10^1	10^2	10^3	10^2	10^2	10^4–10^5
Time to Master for Humans	Seconds	Minutes	Hours	Hours	Hours	Years

BALROG unifies six environments, each targeting a distinct combination of agentic skills:

BabyAI (Chevalier-Boisvert et al., 2019) is a gridworld navigation environment with procedurally generated missions expressed in natural language (e.g., "go to the red ball"). Tasks are solvable in tens of steps and serve as the entry-level component of BALROG, focusing on instruction following and spatial navigation.

TextWorld (Côté et al., 2019) is a text-based adventure game engine generating household tasks (e.g., finding and placing objects). It introduces resource management and requires tracking object states over hundreds of interaction steps, all through natural language without any visual input.

Crafter (Hafner, 2021) is an open-world survival game requiring agents to gather resources, craft tools, and survive over thousands of steps. It evaluates a broad spectrum of skills including exploration, hierarchical planning, and credit assignment through a set of binary achievement milestones.

Baba Is AI (Cloos et al., 2024) is based on the puzzle game *Baba Is You*, in which agents must manipulate both objects and the rules of the game (represented as movable word tiles) to reach a goal. It tests the ability to deduce environment dynamics and plan compositionally under rule uncertainty.

MiniHack (Samvelyan et al., 2021) provides a suite of customizable NetHack-based tasks of varying complexity. We evaluate on five MiniHack sub-tasks spanning navigation, skill use, and quest completion. These environments require understanding of complex game mechanics over hundreds of steps.

NetHack (via the NetHack Learning Environment, NLE (Küttler et al., 2020)) is the most challenging environment in BALROG. A classic roguelike with thousands of item types, complex monster interactions, and procedurally generated dungeons, NetHack demands years of mastery for humans (Hambro et al., 2022a). Solving NetHack requires long-term planning, memory, and strategic reasoning over 10^4–10^5 steps.

Table 1 summarizes the skills, time horizons, and human mastery requirements for each environment.

2.2 Scoring and Metrics

All environments are scored on a 0–100 scale to enable fair comparison across tasks with different structures. For environments with binary success criteria (BabyAI, Baba Is AI, MiniHack sub-tasks), score reflects task completion. For environments with incremental milestones (Crafter, TextWorld, NetHack), score reflects the proportion of achievements or objectives completed per episode.

For NetHack, we introduce a new **data-informed progression metric**. Rather than using raw game score (which is highly skewed and dominated by terminal events), we define progression based on how far through the game's content hierarchy an agent advances, calibrated against the distribution of human and bot trajectories from the NetHack Learning Dataset (Hambro et al., 2022b). This pro-

vides a more semantically meaningful measure of NetHack performance that distinguishes between agents that barely survive the first level and those that explore deeper dungeon levels.

Final reported scores average over multiple random seeds per task to account for procedural variability, with standard errors computed across seeds.

Aggregating performance across environments into a single benchmark score requires careful design. BALROG computes the unweighted mean across all six environments, ensuring that each environment contributes equally regardless of its time horizon or difficulty. This promotes breadth over specialization: a model excelling only at BabyAI navigation while failing at NetHack receives a lower aggregate score than a model achieving moderate progress across all environments. We additionally report per-environment breakdowns so that researchers can identify specific capability gaps. For Crafter, we follow the geometric-mean scoring convention of Hafner (2021) for its achievement-based metric to penalize strategies that cherry-pick a single easy milestone, then convert to our 0–100 linear scale for consistency with the rest of BALROG.

2.3 Observation Spaces

BALROG supports two evaluation modalities:

Language-only: All observations are provided as structured natural-language descriptions. For BabyAI, Crafter, and NetHack/MiniHack, we use custom text wrappers that describe the visible grid, agent inventory, and relevant game state. For TextWorld, the game engine natively produces text output. For Baba Is AI, a custom text wrapper describes visible objects and active rules.

Vision-language: In addition to the text description, models receive the current game frame as an image. For MiniHack and NetHack, this includes ASCII map renderings. For Crafter, pixel-based frames are provided. History observations (beyond the current step) remain text-only in the vision-language setting, which allows us to isolate the effect of visual context on current-step decisions. Future work may extend this to multi-image or video histories.

Both modalities use a consistent history of 16 turns. The initial system prompt includes game rules, a list of valid actions with descriptions, and general instructions formatted in model-agnostic templates compatible with all evaluated APIs.

2.4 Action Space and Error Handling

Models select actions by emitting natural-language strings corresponding to valid actions. On invalid action outputs, BALROG provides feedback to the model ("invalid action: [action]"), logs the error, and executes a safe fallback (e.g., no-op or "move north"). This mirrors realistic agentic deployment where models must handle feedback from environments that do not always accept arbitrary outputs. Invalid action rates are tracked per model and environment as an additional diagnostic.

3 Zero-Shot Evaluation Protocol

3.1 Models Evaluated

We evaluate a broad set of leading closed- and open-source models. For language-only evaluation, we test: **Claude 3.5 Sonnet** (Anthropic, 2024), **GPT-4o** (OpenAI, 2024a), **GPT-4o-mini** (OpenAI, 2024a), **Gemini-1.5-Pro** (Reid et al., 2024), **Gemini-1.5-Flash** (Reid et al., 2024), **Llama-3.1-70B-Instruct**, **Llama-3.1-8B-Instruct** (Dubey et al., 2024), and **Llama-3.2-{1B,3B,11B,90B}-Instruct** (MetaAI, 2024). For vision-language evaluation, we use Claude 3.5 Sonnet, GPT-4o, GPT-4o-mini, Gemini-1.5-Pro, Gemini-1.5-Flash, Llama-3.2-11B-Instruct, and Llama-3.2-90B-Instruct—the subset of models supporting image inputs.

Closed-source models are accessed via their respective APIs (Anthropic, OpenAI, Google). Open-source models are run locally using vLLM for efficient parallel inference. Evaluation is highly parallelized with asynchronous execution to enable large-scale experiments across all environments and seeds.

3.2 PROMPTING STRATEGY

All evaluations are conducted **zero-shot**: no in-context examples are provided. The system prompt describes the game rules, the list of valid actions with brief descriptions, and instructions to respond with a single valid action. This standardized setup ensures fair comparison across models and avoids confounding from example selection. We do not use chain-of-thought prompting in our primary baselines, though BALROG supports optional few-shot in-context learning (ICL) and reasoning traces for future ablations.

3.3 INFRASTRUCTURE

BALROG provides a unified evaluation harness with standardized environment wrappers, client wrappers for OpenAI, Anthropic, and Google Gemini APIs, and vLLM support for local models. The `eval.py` entry point accepts Hydra-based configuration, enabling straightforward sweeps over models, environments, and hyperparameters. Logging captures per-step actions, observations, rewards, and error rates, enabling detailed post-hoc analysis.

4 RESULTS

4.1 LANGUAGE-ONLY PERFORMANCE

Table 2 reports average progress across all BALROG environments for language-only models. Claude 3.5 Sonnet achieves the highest overall score ($32.64 \pm 1.93\%$), closely followed by GPT-4o ($32.34 \pm 1.49\%$). Among open-source models, Llama-3.1-70B-Instruct ($27.88 \pm 1.43\%$) is competitive, while smaller models show a clear performance decline. Gemini-1.5-Pro lags behind the top closed-source models ($21.00 \pm 1.18\%$), and Gemini-1.5-Flash performs even lower ($14.63 \pm 1.37\%$). The smallest models (Llama-3.2-1B at 6.65%) achieve minimal progress, highlighting the importance of model capacity for agentic tasks.

Table 2: Language-Only Performance

Model	Average Progress (%)
claude-3.5-sonnet	32.64 ± 1.93
gpt-4o	32.34 ± 1.49
llama-3.1-70b-it	27.88 ± 1.43
llama-3.2-90B-it	27.29 ± 1.44
gemini-1.5-pro	21.00 ± 1.18
claude-3.5-haiku	19.32 ± 1.83
gpt-4o-mini	17.36 ± 1.35
llama-3.2-11B-it	16.82 ± 1.47
llama-3.1-8b-it	15.14 ± 1.55
gemini-1.5-flash	14.63 ± 1.37
llama-3.2-3B-it	10.13 ± 1.28
llama-3.2-1B-it	6.65 ± 1.04

Table 3: Vision-Language Performance

Model	Average Progress (%)
claude-3.5-sonnet	35.48 ± 2.02
gemini-1.5-pro	25.76 ± 1.36
gpt-4o	22.56 ± 1.44
llama-3.2-90B-it	20.99 ± 1.58
gpt-4o-mini	15.36 ± 1.29
gemini-1.5-flash	14.94 ± 1.40
llama-3.2-11B-it	8.43 ± 1.26

4.2 VISION-LANGUAGE PERFORMANCE

Figure 2 and the right panel of Table 2 show VLM results. Strikingly, adding visual context often *hurts* performance: GPT-4o drops from 32.34% (language-only) to 22.56% in the VLM setting, and Llama-3.2-90B-Instruct falls from 27.29% to 20.99%. The notable exception is Claude 3.5 Sonnet, which *improves* with visual context, rising from 32.64% to 35.48%. Gemini-1.5-Pro also benefits from vision ($21.00\% \rightarrow 25.76\%$). These results suggest that most current VLMs struggle to extract useful information from game frames for decision-making, consistent with findings from Rahmanzadehgervi et al. (2024) on low-level visual reasoning.

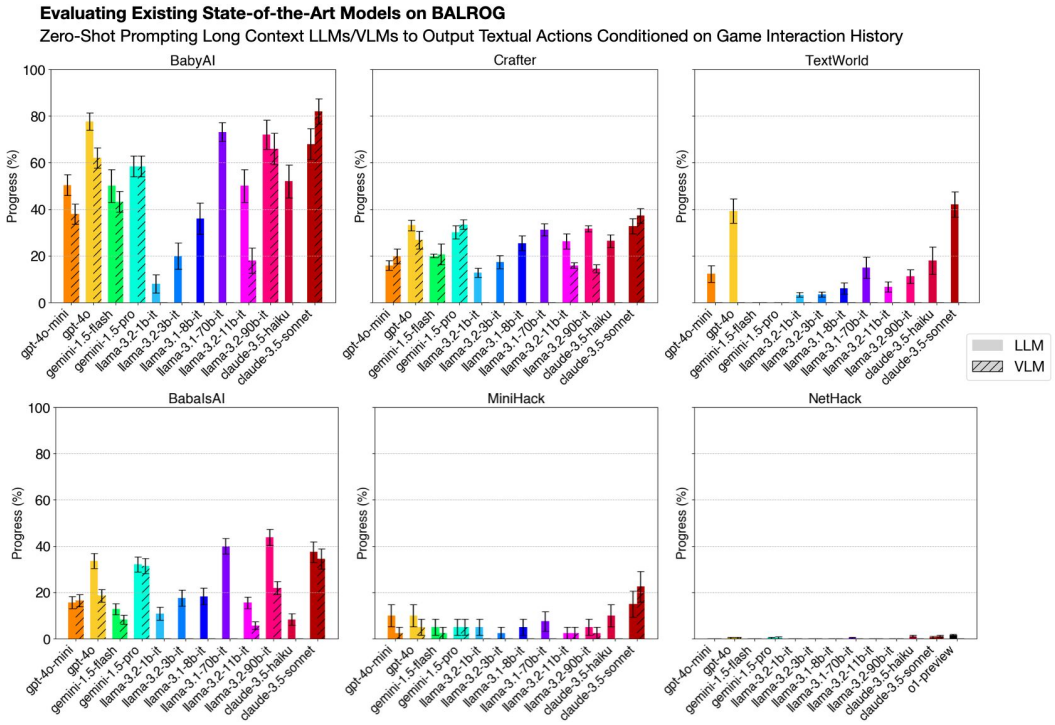

Figure 2: **Baselines for BALROG**. Per-environment and aggregate results comparing language-only (LLM) and vision-language (VLM) settings across all evaluated models. Adding visual observations frequently reduces performance, with Claude 3.5 Sonnet being a notable exception that benefits from vision across environments.

4.3 Per-Environment Analysis

BabyAI is the most tractable environment, with top models solving a large proportion of navigation tasks. Language-only models handle the structured spatial instructions well, consistent with the strong instruction-following capabilities of modern LLMs. However, even in BabyAI, models exhibit unnecessary detours and revisit explored cells, suggesting that spatial memory—not instruction comprehension—is the limiting factor for near-perfect task completion. Success rates across models correlate strongly with instruction-following ability on single-room tasks but diverge sharply on multi-room configurations, where agents must maintain an accurate mental map of previously visited cells and reason about which unexplored corridors remain reachable.

TextWorld reveals an important reliability issue: Gemini-1.5-Pro and Gemini-1.5-Flash both score 0% due to API content filtering that flags the game's adventure text as potentially unsafe, preventing any actions from being executed. Among models that successfully interact with TextWorld, GPT-4o and Claude 3.5 Sonnet lead, demonstrating effective multi-step object tracking and planning. Successful models tend to employ explicit state-tracking strategies, frequently re-reading their current inventory and recalling prior observations before committing to an action. Despite this adaptive behavior, all models plateau after completing roughly half of the available objectives, suggesting that maintaining a complete dependency graph of sub-tasks across a long episode exceeds the effective working memory of current LLM architectures.

Crafter is challenging but partially solved; models make progress on early achievements (gathering wood, collecting resources) but consistently fail at later milestones requiring multi-step crafting chains and long-horizon resource management. Models reliably collect wood and stone but rarely advance to placing furnaces or crafting iron tools, revealing how compounding achievement dependencies create an exponential bottleneck for agents without explicit hierarchical planning over multi-step goal sequences. The vision-language setting yields mixed results in Crafter: while certain models benefit from pixel-based frames for identifying nearby resources and terrain features,

others score lower than their language-only counterparts, apparently distracted by irrelevant visual detail. This asymmetry across models suggests that exploiting visual context in open-world survival games demands not only strong visual recognition but also selective attention to task-relevant features within richly textured scenes.

Baba Is AI proves especially difficult, with Boxoban-style and Quest-type levels remaining entirely unsolved by any model. This suggests current LLMs/VLMs are unable to reason about dynamically mutable game rules and compositional planning under rule uncertainty (Cloos et al., 2024). Analysis of agent trajectories reveals two distinct failure modes: models either attempt to reach the goal object without manipulating any word tiles—ignoring the rule-manipulation aspect entirely—or they push tiles in locally plausible but globally inconsistent sequences that break existing winning conditions without establishing new ones. Both patterns reflect an inability to form and maintain a forward model of how tile configurations determine active rules across multiple steps.

MiniHack is largely unsolved across all models. Even relatively simple MiniHack sub-tasks requiring skill use and navigation in NetHack-like environments expose severe limitations in models' ability to understand complex item interactions and spatial reasoning over ASCII maps. We identify two recurring failure modes: (1) models frequently misinterpret ASCII map symbols, conflating monster glyphs with wall tiles or item tokens with environmental features; and (2) models apply rigid, non-adaptive action strategies that ignore the stochastic outcomes of NetHack-derived mechanics such as combat resolution and item identification. These patterns suggest that MiniHack demands not only spatial comprehension but also probabilistic reasoning about action outcomes under uncertainty.

NetHack remains an extreme challenge. Even the strongest model (o1-preview, OpenAI 2024b) achieves only 1.57% progression under our new metric, with all other models in the 0–1.5% range. This confirms that NetHack's combination of long time horizons (10^4–10^5 steps), procedural diversity, and complex mechanics exposes fundamental limitations in current models' long-term planning (Jeurissen et al., 2024; Hambro et al., 2022a).

4.4 Invalid Action Rates and Model Reliability

Beyond task-completion scores, BALROG records per-step invalid action rates as a diagnostic of model reliability in interactive settings. Smaller open-source models (Llama-3.2-1B and Llama-3.2-3B) produce invalid actions on a substantial fraction of steps—sometimes exceeding 20% in environments with constrained vocabularies such as NetHack and Baba Is AI—while frontier models (Claude 3.5 Sonnet, GPT-4o) remain below 5% invalid actions in most environments. Across all models, the VLM setting slightly increases invalid action rates relative to language-only evaluation, suggesting that attending to image tokens competes with reliable action formatting. The relationship between invalid action rate and task completion is imperfect: several mid-tier models achieve non-trivial progress despite frequent invalid outputs by recovering effectively from environment error feedback, underscoring the importance of robust error handling in the BALROG harness and in agentic deployment more broadly.

4.5 Key Findings

Our experiments reveal four consistent findings:

1. **Vision hurts most models**: Adding visual observations to language-capable models frequently degrades performance, suggesting that most VLMs fail to effectively use game-frame information for sequential decision-making.
2. **Long-horizon planning is the bottleneck**: Even top models saturate on short-horizon environments (BabyAI) while failing on long-horizon tasks (NetHack, MiniHack), confirming that current LLMs/VLMs do not generalize their reasoning to extended action sequences.
3. **A knowing–doing gap persists**: Models that can describe NetHack strategies in prose fail to execute them when acting step-by-step in the game, revealing that parametric knowledge does not translate to effective sequential decision-making (Kambhampati et al., 2024).
4. **Model scale matters but is not sufficient**: Larger models consistently outperform smaller ones, but even the largest models remain far from human-level performance on complex environments.

5 RELATED WORK

LLM/VLM Benchmarks. Static benchmarks such as MMLU (Hendrycks et al., 2020), GPQA (Rein et al., 2023), and LiveBench (White et al., 2024) measure language understanding and reasoning but do not test interactive decision-making. AgentBench (Liu et al., 2023) evaluates LLMs on eight agent tasks including web navigation and coding, but tasks are generally short-horizon. GAIA (Mialon et al., 2023) and SWE-Bench (Jimenez et al., 2023) extend to more complex real-world tasks but still lack procedural generation and long-horizon temporal structure. BALROG complements these by focusing on the sequential, long-horizon regime. WebArena (**?**) and VisualWebArena (**?**) evaluate LLMs on realistic web-based tasks requiring multi-step navigation and interaction, sharing our focus on long-horizon planning but differing in that web environments are not procedurally generated and admit memorization of specific page layouts.

Game-Based Evaluation. Several works have used games to probe LLM capabilities. SmartPlay (Wu et al., 2023) proposes a suite of six games but focuses on shorter horizons. Jeurissen et al. (2024) evaluate LLMs on NetHack specifically, finding similar limitations to ours. Cloos et al. (2024) evaluate VLMs on Baba Is You, noting dramatic failure on rule-manipulation levels. Voyager (Wang et al., 2023) and MineDojo (Fan et al., 2022) use Minecraft as a testbed, with Voyager showing impressive open-ended exploration via code generation. BALROG unifies multiple environments with standardized protocol, enabling fair cross-environment comparison.

RL Environments for LLMs. BabyAI (Chevalier-Boisvert et al., 2019), Crafter (Hafner, 2021), TextWorld (Côté et al., 2019), MiniHack (Samvelyan et al., 2021), and the NetHack Learning Environment (Küttler et al., 2020) have each been used individually to study RL agent capabilities. BALROG is the first to integrate all of these into a unified testbed with a consistent evaluation protocol for LLMs and VLMs.

Agentic Strategies. Research on improving LLM agents has explored chain-of-thought prompting (Wei et al., 2022), tree-of-thought search (Yao et al., 2024), self-reflection (Shinn et al., 2023), and in-context learning (Agarwal et al., 2024; Laskin et al., 2022). BALROG's open-source toolkit is designed to facilitate ablation studies of these strategies across our diverse environment suite.

Vision-Language Models. GPT-4o (OpenAI, 2024a), Gemini 1.5 (Reid et al., 2024), and Claude 3.5 Sonnet (Anthropic, 2024) represent state-of-the-art VLMs. While these models achieve impressive results on static vision benchmarks, Rahmanzadehgervi et al. (2024) show they struggle with low-level spatial reasoning. Our results extend this finding to the interactive decision-making setting, where visual game observations must be integrated into sequential action selection.

6 OPEN RESEARCH PROBLEMS

BALROG reveals several directions for future research:

Long-context integration. Current models use 16-turn histories in BALROG. Scaling context length and developing better mechanisms for compressing and attending to long game histories may be essential for NetHack-level performance. Promising directions include episodic summarization, where key events are condensed into persistent memory representations, and retrieval-augmented approaches that index past observations by semantic relevance to the current game state. The difficulty gradient across BALROG environments—from BabyAI's short episodes to NetHack's multi-thousand-step trajectories—provides a natural testbed for evaluating such long-context architectural innovations.

Visual reasoning for decision-making. The consistent performance drops when adding vision suggest that new training or prompting strategies are needed for VLMs to effectively use pixel-level game observations in sequential decision-making.

Exploration strategies. Most models in BALROG exhibit narrow exploration, revisiting known states rather than systematically mapping unknown territory. Integrating structured exploration strategies (Laskin et al., 2022) with LLM agents is a promising direction.

Memory and state tracking. Long-horizon tasks like NetHack require tracking item identities, dungeon layouts, and character state over thousands of steps. External memory mechanisms and retrieval-augmented approaches (Lewis et al., 2020) may be necessary.

Agentic training. Fine-tuning on game trajectories with reinforcement learning from environment feedback (Ouyang et al., 2022) could substantially improve performance on BALROG tasks, complementing zero-shot prompting.

Hierarchical planning and goal decomposition. The failure of current models on multi-step achievement sequences in Crafter and on complex quests in NetHack points to a deeper limitation: the absence of principled hierarchical planning. While chain-of-thought prompting encourages step-by-step reasoning, it does not provide mechanisms for decomposing long-horizon goals into manageable sub-goals or for tracking progress toward intermediate milestones. Methods that augment LLMs with explicit goal hierarchies, learned skill libraries, or programmatic sub-task planners (Wang et al., 2023) may be necessary for substantial gains on the hardest BALROG environments. Developing and benchmarking such hierarchical agents—and understanding when flat prompting suffices versus when structured decomposition is essential—is an important open research direction that BALROG is well positioned to support.

Knowing–doing gap. The gap between models' stated game knowledge and their in-context action quality calls for new evaluation paradigms that separately assess declarative knowledge and procedural execution (Kambhampati et al., 2024).

Multi-agent collaboration. Distributing the planning burden across multiple cooperating LLM agents—where subagents specialize in inventory management, spatial navigation, or long-term strategy—could alleviate the memory and reasoning bottlenecks exposed by BALROG, particularly in NetHack and MiniHack where the state space is too large for any single context window to adequately represent.

7 Limitations

While BALROG provides a comprehensive testbed for agentic evaluation, several limitations merit acknowledgment. First, all baselines are conducted zero-shot without task-specific fine-tuning; stronger performance may be achievable with in-context examples or domain adaptation, and future work should explore the contribution of these factors systematically. Second, our evaluation relies on a fixed number of random seeds; the true variance of procedurally generated environments may require additional seeds for precise estimation on low-scoring tasks such as NetHack. Third, closed-source models are accessed via APIs, precluding direct inspection of intermediate representations or attention patterns that could help illuminate the knowing–doing gap. Fourth, human baseline performance is rigorously available only for Crafter and NetHack; collecting controlled human baselines across all six environments would strengthen interpretation of the absolute difficulty levels reported and clarify how much headroom remains for future agentic systems.

8 Conclusion

We introduced BALROG, a benchmark for evaluating large language and vision-language models as autonomous agents on long-horizon, procedurally generated game environments. By unifying six diverse environments under a standardized evaluation protocol, BALROG reveals fundamental deficits in current models: even top-tier systems achieve limited progress on complex environments, vision often hurts rather than helps, and NetHack remains essentially unsolved. These findings demonstrate that existing LLM/VLM capabilities do not straightforwardly transfer to the long-horizon, interactive decision-making regime. We hope BALROG will serve as a rigorous, reproducible challenge to drive progress toward genuinely agentic AI systems. The consistent performance gaps across environments—especially in long-horizon settings—underscore that scaling model size and context length alone will likely be insufficient; qualitative improvements in planning, memory, and visual grounding are all needed. By providing a controlled, reproducible testbed with standardized metrics, BALROG offers the community a shared reference point for measuring progress on these challenges over time. Our benchmark, evaluation code, and public leaderboard are available at `https://balrogai.com`.

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